

Predicting Future Forest Fires

GitHub Link: <https://github.com/christinabarton/PredictingWildfires-CS3>

Context:

Climate change has led to a drastic increase in wildfires around the globe in recent years. Research shows that higher temperatures, earlier snowmelt, longer droughts, and drier air have made fire seasons longer and increased the amount of dry vegetation that can burn. This can have a massive impact on the environment, property, and human mortality of an area. More importantly, these wildfires are occurring in areas with high populations, leading to amplified economic losses and human impact. Modeling these fire occurrences over time offers an opportunity to predict when future fires will occur so that we may better prepare for them.

Motivation:

You are a data scientist working with the California fire department that aims to better prepare their residents for wildfires. They have observed certain qualities about the weather over time and are curious if there is a pattern of events over time that can help us understand when a fire will occur. Your goal is to utilize these weather factors and create a machine learning model that can predict when a wildfire will occur and help the California fire department and citizens.

Deliverable:

The student will develop a system that can determine whether or not a wildfire will occur on a specific day in California. To do this, the student will train a Logistic Regression model and a Random forest model to use time series data to make predictions. The final deliverable will include how well the models performed and their predictions for the first few days of 2024.